Q.25 Explain automated optical method of blood cell counting.

Q.26 Mention applications of centrifuge in laboratory.

Q.27 Explain the construction and working of incubator.

Q.28 Describe the principle and uses of autoclave.

Q.29 Explain the importance of optical filters in photo colorimeter.

Q.30 Describe applications of flame photometer in biochemical analysis.

Q.31 Explain different sections of an auto-analyzer.

Q.32 Describe Coulter principle for blood cell counting.

Q.33 Explain the principle and applications of electrophoresis.

Q.34 Write a short note on chromatography.

Q.35 Write applications of oven in microbiology laboratory.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw the block diagram of a spectrophotometer and explain its working.

Q.37 With neat diagram, explain the working principle of a flame photometer.

Q.38 Describe the working and applications of an auto-analyzer with block diagram.

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4th Sem / Medical Electronics

Subject:- Medical Laboratory Instruments/ Analy Inst. (BM)

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Principle of photo colorimeter is based on :

a) Reflection of light

b) Absorption of light by solutions

c) Refraction of light d) Interference of light

Q.2 In a spectrophotometer, diffraction grating is used as:

a) Detector b) Monochromator

c) Printer d) Amplifier

Q.3 The atomizer if flame photometer is used to:

a) Filter the light

b) Convert liquid sample into fine mist

c) Supply LPG d) Record signals

Q.4 pH of blood normally ranges between:

a) 6.2-6.5 b) 7.35-7.45

c) 8.0-8.5 d) 5.5-6.0

Q.5 In auto-analyzer, the section that mixes reagents and sample is:

a) Dialyzer b) Sampler

c) Mixer d) Printer

Q.6 Blood cell counting by electrical conductivity method is called:

a) Coulter principle b) Chromatography

c) Electrophoresis d) Spectrometry

Q.7 The instrument used for separating components by high-speed rotation is :

a) Centrifuge b) Oven

c) Autoclave d) Microscope

Q.8 Which sterilization method uses steam under pressure?

a) Incubator b) Autoclave

c) Chromatography d) Electrophoresis

Q.9 The detector used in spectrophotometer is:

a) Photodiode b) Electrode

c) Atomizer d) Filter

Q.10 Flame photometer is mainly used for measuring:

a) Sodium and potassium

b) Blood proteins

c) Enzymes d) Glucose

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Write any two applications of a photo colorimeter.

Q.12 Name the monochromator used in spectrophotometer.

Q.13 Write the normal value of blood pH.

Q.14 Mention one use of flame photometer in clinical laboratory.

Q.15 Write any two sections of an auto-analyzer.

Q.16 Expand RBC and WBC.

Q.17 Write the principle of centrifuge in one line.

Q.18 Name the sterilization method used for culture media.

Q.19 Mention one application of electrophoresis.

Q.20 Write one use of chromatography.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write specifications of a photo colorimeter.

Q.22 Explain the role of monochromator in spectrophotometer.

Q.23 What is the significance of pH measurement in blood?

Q.24 Explain microscopic method of blood cell counting.